AMD EPYC 9555, 32 threads, scalar matrix

(1) NeoN vs. NeoN Fused

NeoN
NeoN Fused
Face-Based
Global Face-Based
Cell-Based
Batched Face-Based

Matrix Assembly Throughput [GB/s]

0
5
10
15
20
25

1 k
10 k
100 k
1 M
10 M
64 M

(2) NeoN Fused vs. JuNe

NeoN Fused
JuNe
Face-Based
Global Face-Based
Cell-Based
Batched Face-Based

Matrix Assembly Throughput [GB/s]

0
5
10
15
20
25

1 k
10 k
100 k
1 M
10 M
64 M

(3) speedup of JuNe over NeoN Fused of the same kind

Face-Based
Global Face-Based
Cell-Based
Batched Face-Based

Speedup of JuNe over NeoN Fused

2
1.5
1
0.75
0.5
0.3
0.2
0.1
0.05

1 k
10 k
100 k
1 M
10 M
64 M

Cells